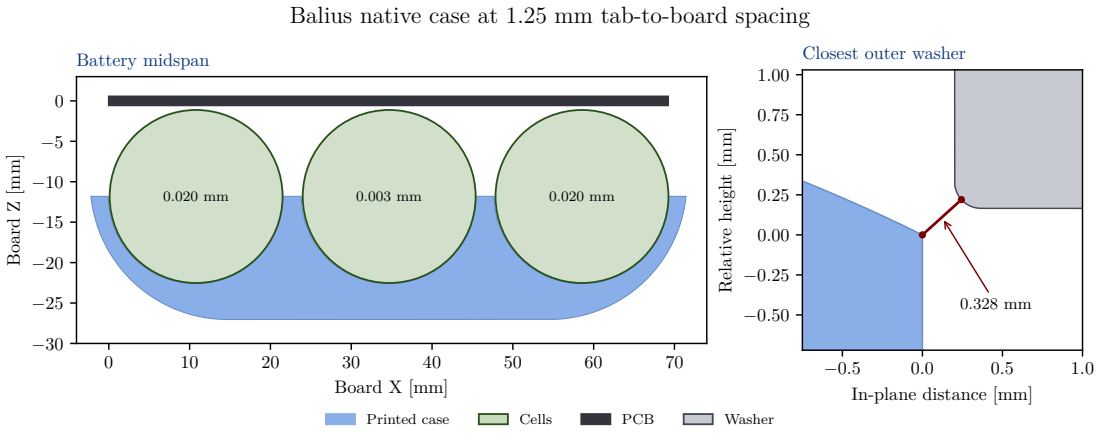
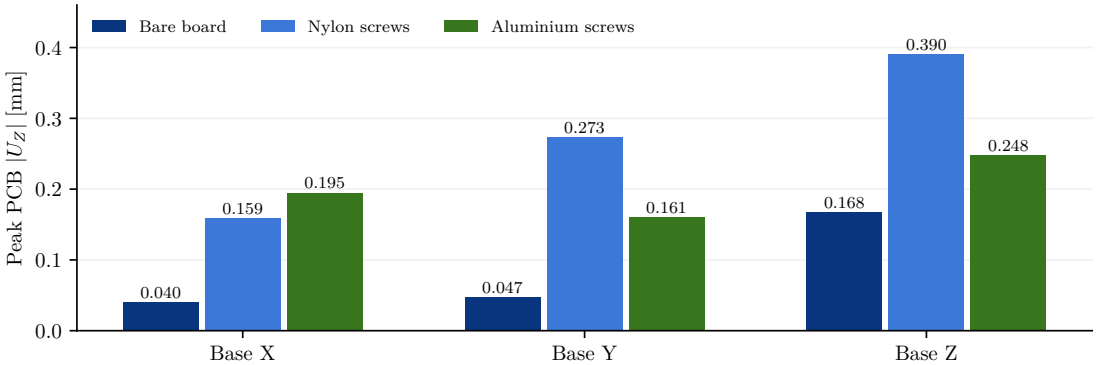
# The three-pocket Balius case fits at 1.25 mm stand-off



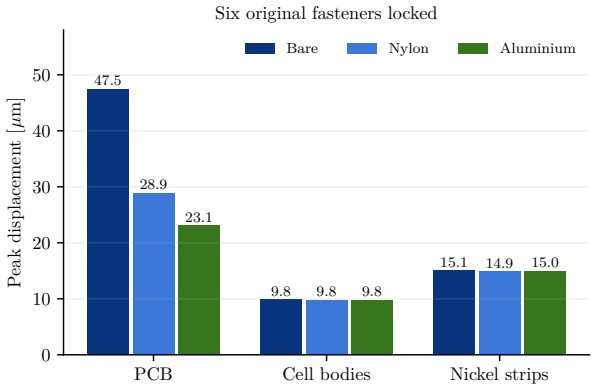

The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. The rigid left cell reaches its pocket after **0.020 mm toward the middle cell** or **0.165 mm away from it**.

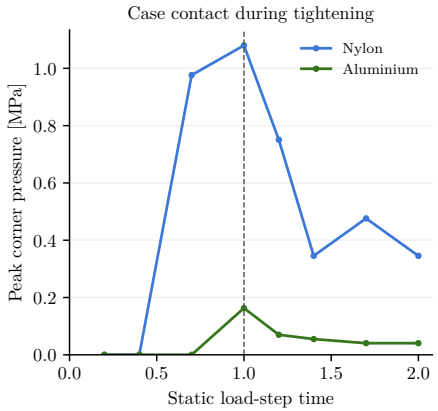
# The snug case increases board travel in the seeded clips




During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

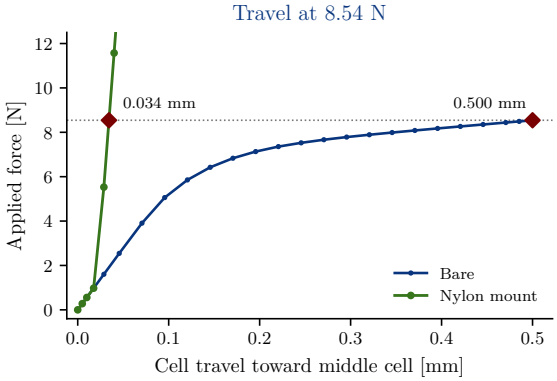
# Both snug mounts reduce PCB movement during tightening

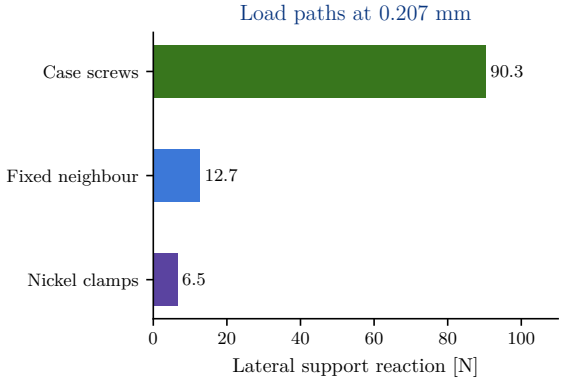



At bolt lock, peak PCB displacement fell from **47.46 $\mu$m bare** to **28.91 with nylon screws** and **23.09 with aluminium screws**. All three nylon meshes put the peak at the same corner, with a spread of 0.055 $\mu$m. Only the middle cell touched the case.

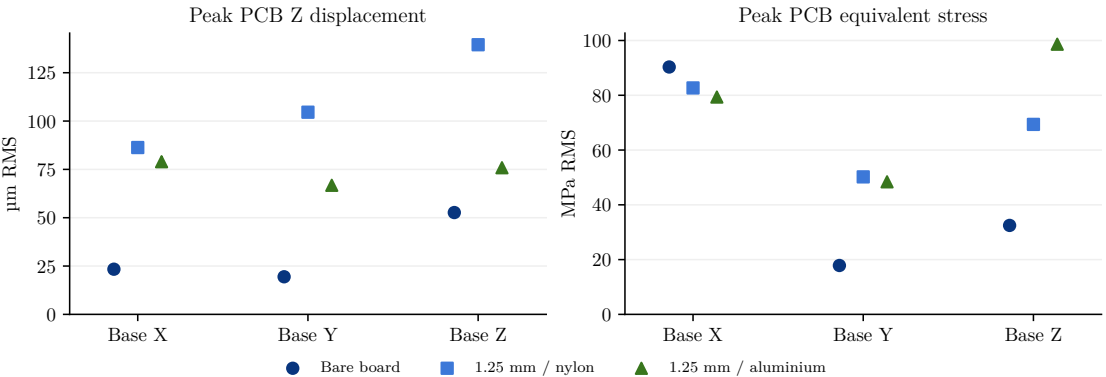
# The snug case reduces sideways cell travel at the same load



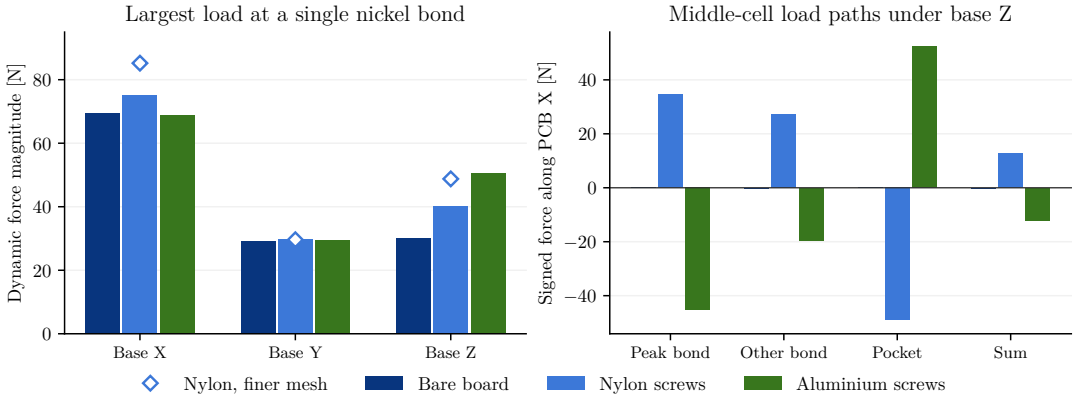

At **8.54 N** toward the middle cell, travel fell from **0.500 mm bare** to **about 0.034 mm with nylon screws**. Bare nickel reached 1.25% plastic strain at that load. Both cased states bracketing it stayed elastic. At 0.207 mm travel the screws carried 90.3 N of 109.5 N. The aluminium pull is queued.

# Aluminium cuts board travel relative to nylon but raises base-Z stress



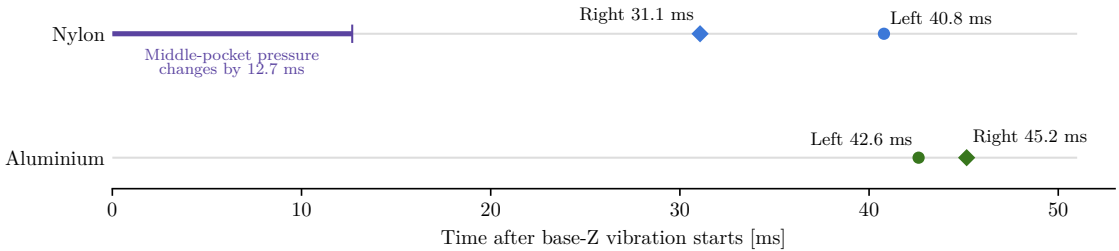

Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

# Board-normal vibration loads the middle cell's nickel bond



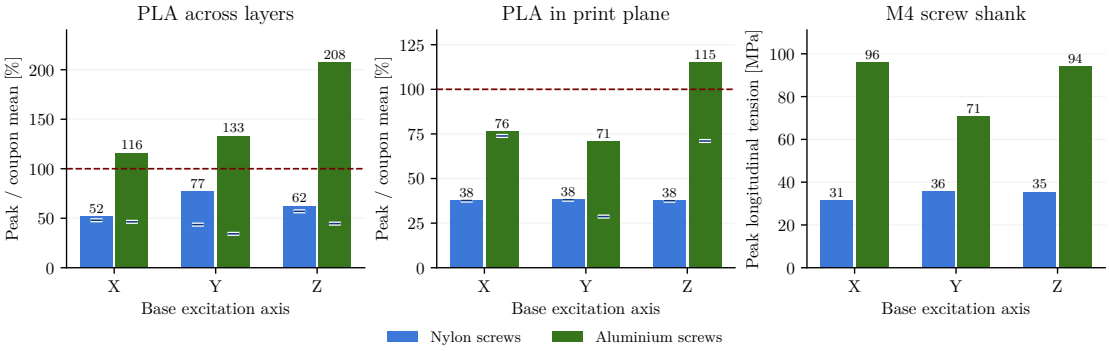

The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The finer nylon mesh gave **48.8 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories at the installed contact state. The diamond uses independent integration on the finer native modes.

# The first outer cell to reach the case changes with the screws



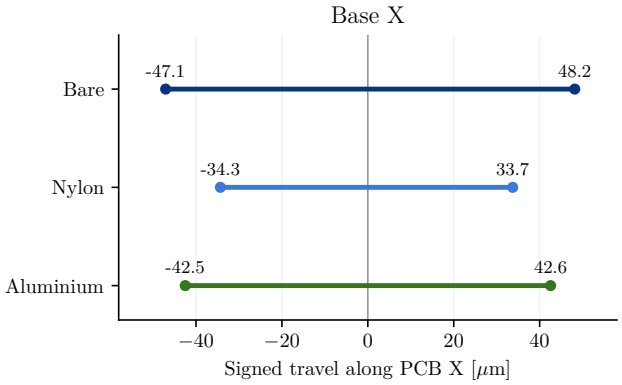

The native base-Z history puts the **right cell first at 31.1 ms with nylon** and the **left cell first at 42.6 ms with aluminium**. Refining the nylon case mesh from 2 to 1.5 mm shifted both independently calculated crossing intervals by only 0.015 ms. With nylon, middle-pocket pressure must redistribute by 12.7 ms, before either predicted crossing. Impact forces remain unresolved.

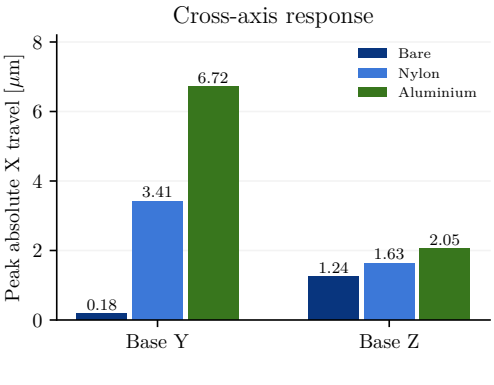
# The printed tab exceeds its coupon reference with aluminium screws




On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.

# The snug case reduces sideways cell travel under base-X vibration




Relative to its two clamps, the left cell's peak lateral travel fell by **29% with nylon screws** and **12% with aluminium**. Both mounts increased the smaller lateral response under base Y and Z. The comparison uses the nine-second native histories with contact fixed at bolt lock.